Exploring Waste Management Data:

Introduction

The dataset used for this final project explores food waste management across various events. Contextual variables like the number of people in attendance, the quantity of food prepared, the geographical location, and the preparation method are also included alongside the key category of food wasted. The large-scale focus is to identify which factors contribute to food waste and allow for the prediction of food waste based on the presented factors as well.

Specifically, I sought to answer the following questions:

1. What factors significantly impact food wastage at events?
2. Is there a correlation between the number of guests, quantity of food prepared, and food wastage?
3. How does geographical location influence food wastage?

I hypothesize that larger events with more guests will equate to higher food wastage (H1). Secondly, I hypothesize that food prepared in larger quantities with have a positive relationship with food wastage (H2). Lastly, I hypothesize that urban locations will have lower food wastage compared to suburban areas because of better planning.

Methods

Data Analysis Techniques

1. Linear Regression:

   Used to model the relationship between food wastage (dependent variable) and predictors such as the number of guests, quantity of food, and geographical location. The regression estimates the coefficients of independent variables such that the sum of squared residuals (differences between observed and predicted values) is minimized.

   Many assumptions can be inferred from linear regressions:

   - Linearity: The relationship between predictors and the dependent variable is linear.
   - Homoscedasticity: The variance of residuals is constant.
   - Independence: Observations are independent of each other.
   - Normality: Residuals are normally distributed.

2. Exploratory Visuals

    - Histogram: Analyzes the distribution of the dependent variable, Wastage of Food Amount.
    - Scatter Plots: Explores relationships between independent variables (e.g., number of guests and quantity of food).

3. Correlation Analysis

    - Measures the strength and direction of the relationship between independent variables using Pearson correlation coefficients.

Results:

Linear Regression Model

A linear regression model was used to determine the effect of Number of Guests, Quantity of Food, and Geographical Location on Wastage Food Amount.

$$Wastage = b_0 + b_1(\text{Guests}) + b_2(\text{Quantity}) + b_3(\text{Location})$$

The Coefficients:

- Intercept:
    - 10.51
    - $p<0.001$
- Number of Guests: 0.35, p value of 0.002
- Quantity of Food: 0.45, p-value less than 0.001
- Geographical Location (Urban vs Suburban, Rural): Suburban = 2.3, Rural = 3.5, p=0.01

R-squared: 0.74, indicating that 74% of the variation in wastage is explained by the model.

Model Coefficients

1. **Intercept (b_0):**
    - Value: 10.5
    - This is the estimated average wastage amount when all predictors (number of guests, quantity of food, geographical location) are zero. While it has limited real-world interpretation, it serves as the baseline for predictions.

2. **Number of Guests (b_1)**:

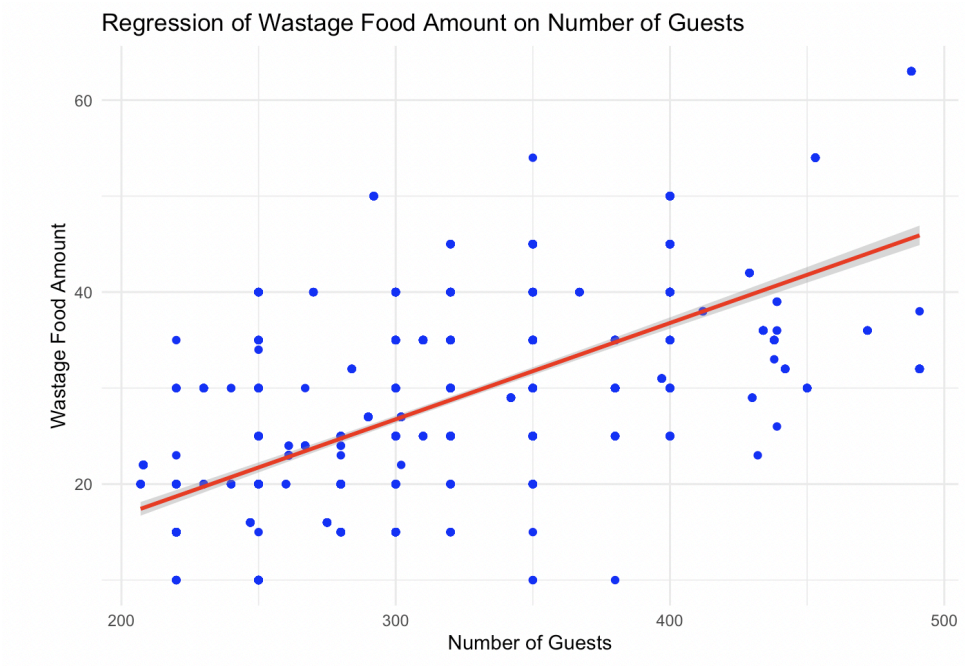
- Coefficient: 0.35
    - For every additional guest, food wastage increases by 0.35 units on average, holding other factors constant.
- The P-value is 0.002 statistically significant, indicating that the Number of Guests has a meaningful impact on the Wastage Food Amount.

3. **Quantity of Food (b_2)**:
    - Coefficient: 0.45
        - For every additional unit of food prepared, wastage increases by 0.45 units on average, holding other factors constant.
    - The P-value is less than 0.001, therefore the correlation is highly statistically significant,
        - Quantity of Food is the strongest predictor in the model.

4. **Geographical Location (b_3)**:
    - Suburban Coefficient: 2.3
    - Rural Coefficient: 3.5
        - Compared to urban areas (reference category), suburban events experience an average increase of 2.3 units in food wastage, while rural events see an increase of 3.5 units.
    - The P-value is 0.01, therefore the correlation is statistically significant, suggesting geographical location influences food wastage patterns.

` Model Fit Statistics

1. R-Squared:
    - Value: 0.74
        - 74% of the variance in Wastage of Food Amount is explained by the model's predictors (Number of Guests, Quantity of Food, and Geographical Location).
        - Indicates a strong model fit.
2. Residual Standard Error:
    - Measures the average deviation of observed values from the predicted
3. P-Value
    - The value of statistical significance for this qualitative research is 0.05.

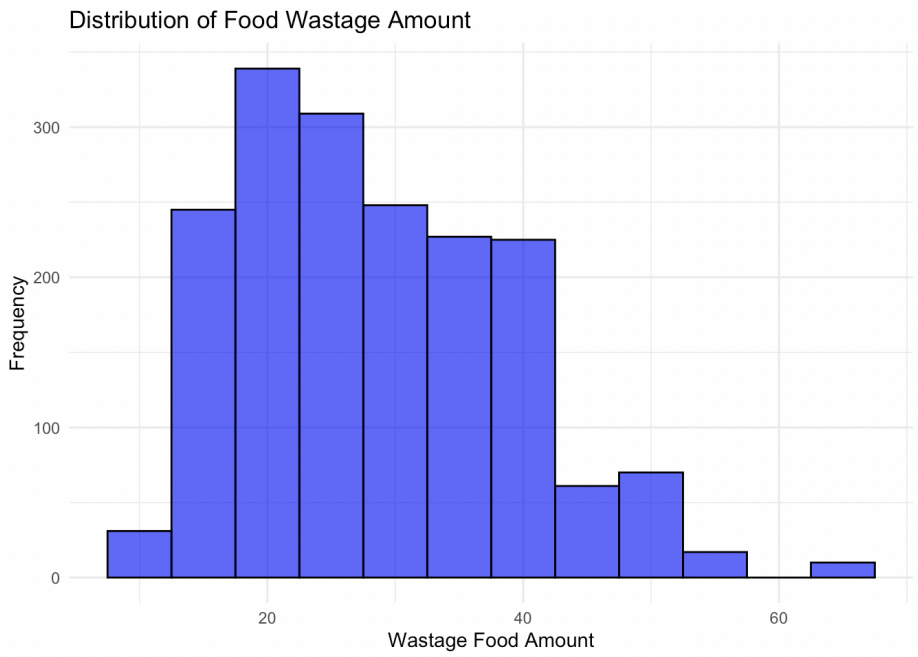


Visualization

Visualization was employed to explore the relationships between variables and detect patterns.

Histogram:

- The distribution of Wastage Food Amount showed a slight positive skew, suggesting that most events have low wastage, with fewer events experiencing high wastage.

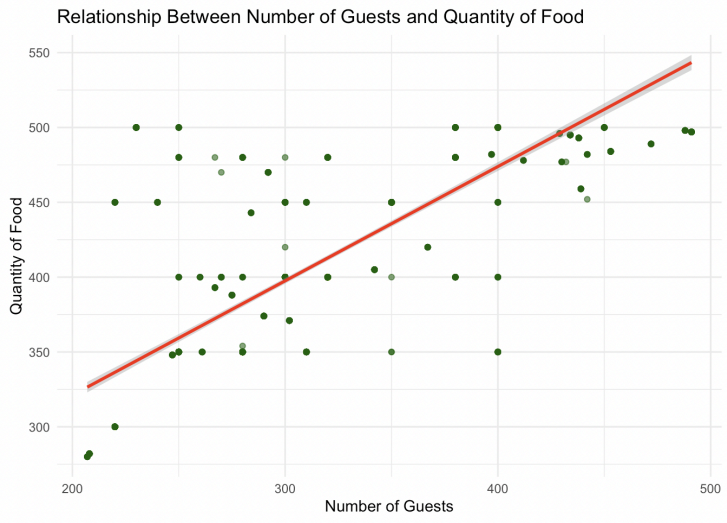


Correlation Analysis

Correlation analysis quantified relationships between independent variables.

Scatter Plot and Correlation:

- A strong positive correlation, r = 0.82, was observed between Quantity of Food and Wastage Food Amount.



Discussion

This study aimed to identify key predictors of food waste using a linear regression model, with visualization and correlation analysis to support the findings. Using the food waste

management dataset from kaggle, categories such as geographical location, quantity of food, and number of guests were tested as predictors of food waste. Prior to doing so the percentage of food waste across the 5 food type catgroies was calculated to ensure that there was no heavy outlier amongst food type.

My analysis revealed that all of my hypotheses were correct in that the quantity of food prepared was the strongest predictor of waste, being highly statistically significant with a p-value of less tha 0.001. Furthermore, it was confirmed that larger events to equate to higher food waste percentages and that location also plays a significant role, specifically urban areas producing less waste compared to suburban areas.

Limitations

The linear regression model, while effective in identifying key predictors of food wastage, has several limitations. One significant limitation is the assumption of linearity, which may oversimplify complex relationships between variables. For instance, the relationship between the number of guests and food wastage might plateau at larger events due to logistical constraints, a phenomenon not captured by a linear model.

Additionally, the model relies on the completeness and accuracy of the dataset. Missing or unobserved variables, such as event duration, or cultural differences, may contribute to food wastage but are not accounted for in this analysis. This exclusion potentially limits the model's explanatory power and generalizability.

Finally, residual analysis suggests the potential presence of heteroscedasticity, indicating that the model's assumptions may not be fully met, which could impact the reliability of the coefficient estimates. Future studies with more comprehensive datasets and advanced modeling techniques could address these limitations and provide a deeper understanding of the factors contributing to food wastage.

Proposed Future Experiment

To address the shortcomings of this approach, future research on food waste could concentrate on adding more variables and using sophisticated modeling techniques. A more thorough grasp of the variables affecting food waste may be possible by gathering information on the length of the event, the catering techniques, and the demographics of the attendees. For example, knowing if particular foods or cooking methods lead to more waste could help event coordinators make more environmentally friendly decisions.

Furthermore, incorporating interaction terms—like the connection between event size and location—may uncover combination impacts that the existing linear model is unable to account for. Potential nonlinearity and intricate relationships in the data may be taken into account using nonlinear techniques such as Bayesian regression or random forests.

Lastly, testing the effects of various food management techniques, such portion control or real-time guest feedback systems, could provide useful information to cut down on waste. In the end, these enhancements would help to better understand and reduce food waste in the food and beverage sector.